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Total No. of printed pages = 03

Winter 2025

B. Tech. 4th Semester Examination
Construction Engineering and Management

Course Code: BCE23406T

Full Marks – 50

Time – 2 hours

The figure in the margin indicates full marks for the questions.

1. Choose the correct answer from the choices given:

(10 × 1=10)

(i) Identify the option that does not represent a type of construction project.

(a) Residential

(b) Commercial

(c) Industrial

~~(d) Shallow foundation~~

(ii) Which of the following represents a dummy activity?

~~(a) Artificially introduced~~

(b) represented by a straight line

(c) Well defined activity

(d) none of the above

(iii) The latest finish time of an activity is the

(a) T_E of the tail event

(b) T_E of the head event

(c) T_L of the tail event

~~(d) T_L of the head event~~

(iv) Which of the following is not a type of shallow foundation?

(a) Spread footing

~~(b) Pile foundation~~

(c) Combined footing

(d) Strap footing

(v) What is the primary purpose of formwork in RCC?

(a) to reduce the weight of the structure

(b) to enhance the appearance of concrete

~~(c) to shape and support concrete until it gains strength~~

(d) to prevent corrosion of steel bars

(vi) Which construction equipment is best suited for earthmoving?

~~(a) Bulldozers~~

(b) Pumps

(c) Concrete mixers

(d) Static cranes

(vii) What is the main goal of lean construction?

- (a) to increase project complexity
- ☒ (b) to minimize waste and maximise value
- (c) to delay construction for design perfection
- (d) to use more labor in more time

(viii) Long term exposure to airborne contaminants and dust can lead to which health issue?

- (a) Hearing loss
- (b) skin rash
- (c) heat stroke
- ☒ (d) Asthma

(ix) The purpose of the force majeure clause in a construction contract is to

- (a) ensure subcontractors are paid on time
- (b) hold the contractor liable for all damages
- ☒ (c) free parties from liabilities due to an unforeseen event
- (d) specify the contract's start and end dates

(x) Identify the option that does not belong to the recognized phases of a construction project

- (a) Project initiation
- ☒ (b) Project manager
- (c) Project execution
- (d) Project closure

2. Answer any *one* question from the *two* questions provided.

- ☒ (a) What do you mean by the term 'Construction Project Management'? Discuss about the agencies involved in CPM and their methods of execution. (4+6=10)

OR

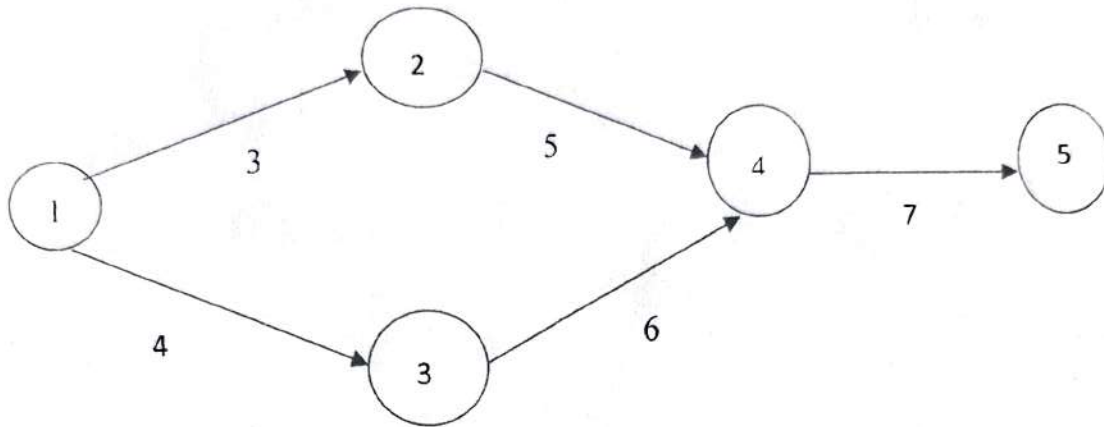
- (b) Explain about the features of a construction project. (10)

3. Answer any *one* question from the *two* questions provided.

- ☒ (a) Identify and classify the different types of activities used in project network. Define Work break-down structure and list its 5 major levels. (5+5=10)

OR

- (b) Determine the earliest start time, earliest finish time, latest start time, latest finish time, total float, free float and independent float in respect of all the activities of the network. Find project completion time and critical path also. (10)



4. Answer any *one* question from the *two* questions provided.

(a) Identify and classify the different types of foundations used in construction with appropriate diagrams.

Discuss the importance of construction equipments in modern construction projects. (6+4=10)

OR

(b) Define formwork and list the different types of formwork. Compare conventional construction methods and mechanized construction methods in terms of labor involvement, speed, quality and cost. (5+5=10)

5. Answer any *one* question from the *two* questions provided.

(a) What do you mean by BIM? Discuss the occupational health hazards faced by construction workers.

(5+5=10)

OR

(b) Identify the causes of accidents at construction sites, explain its effects and recommend suitable preventive measures. (10)